A direct small-noise confinement step

Independent bounded research note

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1 The small-noise half-amplitude step without more cells

Fix an actual integer $n \geq 42$. Let (P, H) be a fixed-target maximizing pair with nonnegative excess $\mathfrak{a}$ over $c = C_E$, and assume the true predecessor bound $C_{n-1} \leq c$. Retain the closed-support contact theorem. Suppose the already acquired panel bounds give

$$p = \mathbb{P}(X > 0) \in [41/100, 42/100], \quad X = \ell(Y - p), \quad Y = B + U, \quad B = \mathbf{1}_{X>0}, \quad \mathbb{E}(U \mid B) = 0,$$

$$|U| < 3/5, \qquad U_0 \leq d < 69/1000, \qquad U_1 \geq -d, \qquad \lambda = n\mathbb{E}U^2 \leq 3/40.$$

Here X is centered with variance one and $\ell > 0$. In particular the initial strip is established before this argument uses bounded conditional moments. These are statements about one actual marginal; no relaxation is assumed to be a probability law.

Theorem 1. *Under these hypotheses $|U| < 1/2$. The conclusion holds for every raw threshold $H \in \mathbb{R}$, both discrepancy orientations after their proper closed reflection, and every actual index $n \geq 42$. It includes $\lambda = 0$ without taking a positive-noise limit.*

1.1 Moment bounds that vanish with the actual noise

Write $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2 = \lambda/n$ and $\theta = \mathbb{E}(X - \mathbb{E}[X \mid B])^2$. Conditional centering and variance give

$$\ell = (v + s)^{-1/2}, \quad \theta = \frac{s}{v + s} = \frac{\lambda}{nv + \lambda}, \quad m^2 = v(1 - \theta), \quad m = \mathbb{E}X_+.$$

Put $v_- = 2419/10000$, $v_+ = 609/2500$ and $\Theta = 733/100000$. Then

$$\theta \leq \frac{3/40}{42v_- + 3/40} < \Theta, \qquad \ell > \frac{1009}{500} = 2.018,$$

where the latter follows by the exact square comparison $(1009/500)^2 v_+ < 1 - \Theta$.

The sign of $B - p + U$ is the actual sign used to define B. Expanding the absolute cubic on these two actual sign classes, and using conditional centering, yields the exact identity

$$\rho = \ell^3 \left\{ v(1 - 2v) + 3\mathbb{E}[|B - p|U^2] + \mathbb{E}[\operatorname{sgn}(B - p)U^3] \right\}.$$

Since $|U| \leq 3/5$ and $p < 1/2$,

$$\rho \leq \rho_B(p) + \frac{3(1 - p) + 3/5}{v^{3/2}} \frac{\lambda}{n}, \qquad \rho_B(p) = \frac{1 - 2pq}{\sqrt{pq}}.$$

The function ρ_B decreases on this panel. The exact comparisons

$$24^2 v_-^3 > \{3(1 - 41/100) + 3/5\}^2, \qquad (21/20)^2 v_- > (1 - 2v_-)^2$$

therefore imply

$$\rho < \frac{21}{20} + 24\frac{3/40}{42} = \frac{21}{20} + \frac{3}{70} < \frac{1093}{1000} =: r. \tag{1}$$

This proves the coefficient24 from the previously acquired strip; it does not assume the half-amplitude conclusion.

Let $\theta_{\pm}$ be the two unnormalized conditional variance contributions. They are nonnegative and sum to θ. Consequently

$$\beta = \mathbb{E}[X|X|] = 1 - 2p - 2q\theta + 2\theta_{+}, \quad 1 - 2p - 2q\theta \le \beta \le 1 - 2p + 2p\theta.$$

Both bounding functions, with θ replaced outward by Θ, decrease in p. Thus

$$\frac{151}{1000} < 1 - 2(42/100) - 2(58/100)\Theta \le \beta \le 1 - 2(41/100) + 2(41/100)\Theta < \frac{187}{1000}. \tag{2}$$

No assertion of uniform integrability, optimizer structure, or independent conditional noise laws is used in these single-marginal moment identities.

1.2 Uniform Gaussian contact bounds at the new boundaries

At every point t of the closed support, fixed-target contact gives

$$0 \le \mathcal{H}_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - n\mathfrak{a}, \qquad e_n = n(\sqrt{n/(n-1)} - 1) - \tfrac{1}{2}. \tag{3}$$

Set $v_n = n - 1$, $s_n = \sqrt{n/v_n}$ and $F_v(x) = \Phi(x/\sqrt{v})$. The exact heat expression is

$$\frac{\mathcal{H}_n(H,t)}{n^{3/2}} = F_{v_n}(H - t) - F_n(H) + tF_n'(H) - \frac{t^2 - 1}{2}F_n''(H).$$

Expand the spatial term through degree four. In the other three terms, expand in the variance variable to orders two, one, one, using $\partial_v F_v = F_v''/2$. After cancellation,

$$\frac{\mathcal{H}_n(H, \pm u)}{s_n^3} = \pm\frac{u^3 - 3u}{6}(1 - z^2)\phi(z) + \frac{u^4 - 6u^2 + 3}{24\sqrt{v_n}}\phi'''(z) + R, \quad z = H/\sqrt{v_n}.$$

The four Taylor remainders are bounded by

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120 v_n}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v_n}, \quad \frac{\|\phi^{(5)}\|_\infty}{48 v_n^{3/2}}, \quad \frac{|u^2 - 1|\|\phi^{(5)}\|_\infty}{16 v_n^{3/2}}.$$

Fourier inversion of the Gaussian gives $\|\phi''\|_\infty \le \phi(0)$, $\|\phi'''\|_\infty \le 2/\pi < 2\phi(0)$, $\|\phi^{(4)}\|_\infty \le 3\phi(0) < 6/5$, and $\|\phi^{(5)}\|_\infty \le 8/\pi < 51/20$. These follow by integrating the absolute powers of frequency; the last comparison uses $\pi > 160/51$. Thus for every $u \ge 1$,

$$|R| \le \frac{u^5/100 + 3u/20}{v_n} + \frac{51(3u^2 - 2)}{960 v_n^{3/2}}.$$

This is the precise domain extension: the remainder requires only $u \ge 1$, and the positive leading coefficient requires $u > \sqrt{3}$. The sign of $u^4 - 6u^2 + 3$ is handled by its absolute value.

For $n \ge 42$, $s_n < 81/80 =: S$, $\sqrt{v_n} > 6$ and $e_n < 8/875$. For the last bound, $s_n < 101213/100000$ by squaring, and the increasing function $s^2/(s+1) - 1/2$ is less than $8/875$ at that rational upper. Set $c_- = 819/2000 < c$ and

$$\begin{aligned} E(u) &= u^5/4100 + 3u/820 + 17(3u^2 - 2)/78720, \\ Q_{r,b}(u) &= -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r, \\ P_{r,b}(u) &= \tfrac{2}{5}S^3\left\{\frac{u^3 - 3u}{6} + \frac{|u^4 - 6u^2 + 3|}{72}\right\} + S^3E(u) + c_-Q_{r,b}(u). \end{aligned}$$

At a boundary with $Q_{r,b} < 0$, this is an upper bound for the right side of (3), after dropping $-n\mathfrak{a} \le 0$. For the negative ray use $b = -151/1000$, and for the positive ray $b = 187/1000$, by (2). The multiplier substitution is made only after checking the sign of Q.

Here the two rational boundaries are

$$u_- = 459/250 = 1.836, \qquad u_+ = 2179/1000 = 2.179.$$

Both exceed $\sqrt{3}$. Independent exact arithmetic verifies

$$P_{r,-151/1000}(u_-) < 0, \qquad P_{r,187/1000}(u_+) < 0.$$

For full rays, the derivative is bounded uniformly in H by

$$|\partial_t \mathcal{H}_n(H,t)| \le \phi(0)\{t^2/2 + 1/2 + e_n\}.$$

Indeed separate the second-order spatial remainder of the normal density at variance n, bounded by $\phi(0)t^2/(2n^{3/2})$, from its variance change. The heat integral for the latter, multiplied by $n^{3/2}$, is at most $\phi(0)n(s_n - 1) = \phi(0)(1/2 + e_n)$. At each boundary the checker verifies the six strict signs

$$Q, Q', Q'', L, L', L'' < 0, \qquad L(u) = \tfrac{2}{5}(u^2/2 + 1/2 + 8/875) + c_- Q'(u).$$

Since $Q''' = -6$ and $L'' = 2/5 - 6c_- < 0$, these signs persist forever. With the moment caps fixed, the contact expression decreases on each ray from its strictly negative boundary. Hence the entire closed rays $(-\infty, -u_-]$ and $[u_+, \infty)$ are absent from the support.

1.3 Amplitude and the closed seam

Finally,

$$u_- < \frac{1009}{500}(41/100 + 1/2), \qquad u_+ < \frac{1009}{500}(3/2 - 42/100).$$

Therefore $-1/2 < Y < 3/2$ throughout the support. Together with the previously acquired inward bounds $U_0 \le d < 0.069$ and $U_1 \ge -d$, this gives $|U| < 1/2$. If $\lambda = 0$, conditional variances are zero and $U = 0$ almost surely directly; the proof never divides by λ.

To join this lemma to the existing eight equal noise bands, use it on the entire closed band $[0, 3/40]$, for every H. Retain the separately verified contact/resource gates on the other seven closed bands covering $[3/40, 3/5]$. Their shared endpoint $3/40$ is included in both covers. The lemma does not assert those other gates pass; their complete source binding and finite inequalities remain separate premises. No added noise or phase cell is required for this alternative composition.